Supplemental Material for "Lattice refinement followed by the rodeo algorithm: a gate-level benchmark of eigenstate-preparation routes on the Hubbard chain"

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This Supplemental Material collects the verification material behind the main text: the sufficient adiabatic time and the response norm of the path (Sec. S1), the dimer-seeded hierarchy and the composition of steps (Sec. S2), the plaquette spectra (Sec. S3), the variational tables (Sec. S4), the Trotterized refinement at eight sites (Sec. S5), the exact-ensemble filter (Sec. S6), the ingredients and validation of the thirty-two-qubit study (Sec. S7), the operating-point tables (Sec. S8), and the convergence of the operating-point search behind Tables 2 and 3 of the main text (Sec. S9). Equation, section, table and figure numbers without the prefix S refer to the main text; all numbers are outputs of the notebook `notebooks/ResolutionRefHubbard.ipynb`.

S1 The sufficient adiabatic time and the response norm of the path

Table S1 evaluates the sufficient adiabatic time of Jansen, Ruskai and Seiler [1] in the form used in paper I, $T_b = \epsilon^{-1/2} \int_0^1 [\|\ddot{H}\|/\Delta^2 + 7\|\dot{H}\|^2/\Delta^3]\, ds$ with $\|B - A\|$ the spectral norm of $(H_{\rm high} - \mu) - P(H_{\rm low} - \mu)P^\dagger$ in the sector, once with the sector gap and once with the gap in the symmetry sector of the tracked state (Fig. 3), against the empirical T^*. The symmetry-resolved bound is smaller by a factor 5 to 17 but remains three to five orders of magnitude above $T^* = 10/t$, as the pairing bound did (there 3×10^2); the larger ratio here reflects $\|B - A\| = 11$–20, set by the shift μ and the width of the Hubbard spectrum. For a rank-one tracked branch the squared projector distance of the bound is the infidelity, so the two criteria measure the same quantity; the slack comes from the conservative norm estimates ($\|B - A\|$ and the global gap where the dynamics pays only for the couplings of the tracked state) and from comparing $\epsilon = 10^{-2}$ with the empirical 10^{-3}, not from an error in either. The table is a numerical quadrature of a rigorous bound, not a certified enclosure of all discretization errors.

The first-order response norm of the path, $\chi_u(s) = \sum_{n>0} |\langle n(s)|\partial_u H|0(s)\rangle|^2/[E_n(s) - E_0(s)]^2$ with $u = \sin^2(\pi s/2)$ and $\partial_u H = (H_{\rm high} - \mu) - P(H_{\rm low} - \mu)P^\dagger$, is the squared eigenvector response (quantum metric) of the tracked state along the affine path in u; it is a sensitivity diagnostic, not by itself the leakage, which also depends on the schedule $u'(s)$ (zero at both ends, so an endpoint maximum of χ_u is not a maximum of the eigenvector speed) and on the oscillatory phases set by T times the integrated gaps. For the pairing model paper I obtained it in closed form at the start of the path, $\chi_u(0) = \|QH_{\rm high}P\psi_{\rm low}\|^2/\mu_{\rm buf}^2$. Figure S1 (left) follows it along Path A for the chain $4 \to 8$: $\chi_u(0) = 0.375$,

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Table S1: Sufficient adiabatic time T_b (infidelity $\epsilon = 10^{-2}$) along Path A ($\mu_{\rm buf} = 2t$) for the chain $4 \to 8$, evaluated with the sector gap and with the gap in the symmetry sector of the tracked state, against the empirical T^* ($1 - p < 10^{-3}$). Times in units of $1/t$.

U/t	$\min \Delta$	$\min \Delta_{\rm eff}$	$\|B - A\|$	$T_b(\Delta)$	$T_b(\Delta_{\rm eff})$	T^*
0	0.532	0.879	11.1	5.7×10^4	1.2×10^4	10
2	0.319	0.780	11.8	2.5×10^5	2.0×10^4	10
4	0.223	0.588	13.9	1.0×10^6	6.4×10^4	10
8	0.137	0.366	20.0	1.0×10^7	5.8×10^5	10

0.527, 0.715 and 0.874 for $U/t = 0, 2, 4, 8$, the maximum sits at the initial endpoint for every coupling, and the norm falls by a factor two to four along the path (at $U = 8t$ it rises again toward $s = 1$, to 0.45, as $\Delta_{\rm eff}$ shrinks). The shift fixes the initial value as it did for the pairing model, and the growth of $\chi_u(0)$ with U is modest, consistent with (not an explanation of) the T^* that does not change with U over the tested range. The right panel compares, with exact slices, the ordinary adiabatic ramp from the free-fermion determinant with the two refinement paths at $U = 4t$: the ramp reaches $p = 0.96878$, 0.99686, 0.99960 and 0.99989 at $T = 4, 8, 12, 16$ (and 0.99999 at $24/t$).

S2 The dimer-seeded hierarchy and the composition of steps

Table S2 shows the first step of the one-dimensional hierarchy, from the filled dimer ($N_\uparrow = N_\downarrow = 2$, one state, $E_{\rm low} = U - 4t$) to the four-site chain: the doubly-occupied seed is a worse start as U grows, $p_0 = 0.80$ at $U = 0$ down to 0.48 at $U = 8t$, but the effective gap is large ($\Delta_{\rm eff} = 2.2t$–$0.8t$) and $p > 0.9997$ is reached by $T = 16/t$ at every U. Figure S2 composes the steps (Sec. 4.3): with the refined four-site state as the input of the $4 \to 8$ step and the shift referenced to its variational energy $\langle\psi|H_{\rm low}|\psi\rangle$, the chain $2 \to 4 \to 8$ at $U = 4t$ with $T = 16/t$ per step reaches $p = 0.99988$ against 0.99994 from the exact coarse input; with $T = 8/t$ in the first step the second saturates at $p = 0.9979$, the overlap of its input. The single-shot refinement $2 \to 8$ with blocks of four fine sites starts from $p_0 = 0.579$ and reaches $p = 0.9966$ at $T = 16/t$ and 0.99988 only at $T = 32/t$.

The step $8 \to 16$ at $(2, 2)$, $U = 4t$ ($784 \to 14\,400$ states) starts from $p_0 = 0.869$ with $E_{\rm low} = -6.817213$ and $E_{\rm high} = -7.209813$. Among the forty lowest eigenstates of the fine sector the first to carry weight lies at $\Delta_{\rm eff} = 0.138t$ (weight 10^{-4}), the next at $0.415t$ and $0.505t$ (the latter with weight 10^{-2}), while the eight other states below $0.4t$, including the one that defines the sector gap ($0.051t$), carry none. With $\mu_{\rm buf} = 2t$ the overlap reaches 0.9924, 0.99931 and 0.99995 at $T = 8, 16, 32$; composing three steps from the filled dimer, $2 \to 4 \to 8 \to 16$ with $T = 16/t$ each and the refined state as the input of every step, gives $p = 0.99923$.

S3 The plaquette

Table S3 collects the plaquette refinements of Sec. 4.4. At $(2, 2)$ the fine ground state has weight 0 on the prolonged coarse ground state (B_1 singlet), 0.647 on the prolonged A_2 triplet, and 0.007 on all other coarse states together ($U = 4t$); the lowest state of the $(A_2, S = 1)$ sector that carries weight lies $1.77t$ above the ground state (among the thirty lowest eigenstates). The $U = 0$ point is degenerate in both models and its labels are not meaningful.

S4 The variational inputs

Table S4 collects the single-layer UCCSD results. On the coarse chain the 26-parameter ansatz is exact at $U = 0$ and reaches infidelities 1.8×10^{-4}, 3.3×10^{-3} and 1.7×10^{-2} at $U/t = 2, 4, 8$ from a free-fermion reference that is very poor at strong coupling ($E_{\rm HF} - E_{\rm FCI} = 2.3t$ at $U = 8t$); on the fine chain the 198-parameter ansatz does slightly better, 4.7×10^{-4}, 3.1×10^{-3}, 1.2×10^{-2}, at 2.3×10^4 CNOTs against 1.5×10^3. Used as the coarse solver, the UCCSD state is prolonged with essentially the same p_0 as the exact coarse state but the refinement then saturates at $p = 0.99669$ ($U = 4t$) and 0.98234 ($U = 8t$) instead of 0.9999. At $U = 4t$, 72 of the 198 fine-chain amplitudes exceed 10^{-2} and 98 exceed 10^{-3}; keeping those 98 (12 singles, 86 doubles) and re-optimizing from

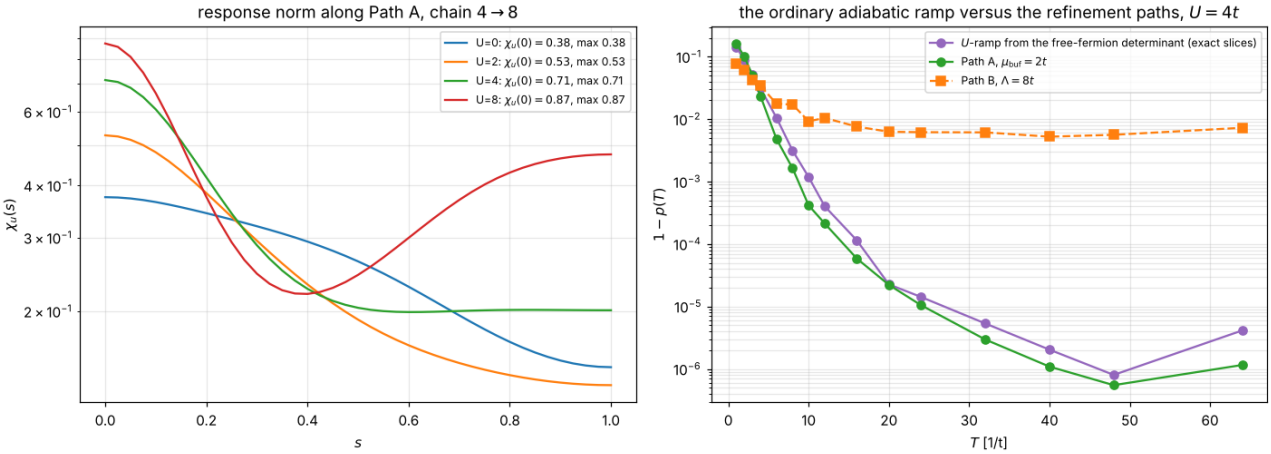


Figure S1: Left: the first-order response norm $\chi_u(s)$ along Path A ($\mu_{\mathrm{buf}} = 2t$) for the chain $4 \to 8$ at $U/t = 0, 2, 4, 8$; the maximum sits at the initial endpoint for every coupling. Right: $1-p(T)$ with exact slices for the ordinary adiabatic ramp from the free-fermion determinant, for Path A and for Path B ($\Lambda = 8t$) at $U = 4t$.

Table S2: The first step of the one-dimensional hierarchy: the filled dimer ($N_\uparrow = N_\downarrow = 2$, one state) refined to the four-site open chain (36 states), Path A with $\mu_{\mathrm{buf}} = 2t$.

U/t	E_{low}	E_{high}	p_0	Δ	Δ_{eff}	$p(8)$	$p(16)$
0	-4	-4.4721	0.805	1.236	2.236	0.9999	1.0000
2	-2	-2.8759	0.754	0.789	1.851	0.9995	1.0000
4	0	-1.9531	0.645	0.540	1.358	0.9979	0.9999
8	4	-1.1172	0.480	0.311	0.805	0.9787	0.9997

the converged values gives $E = -5.618755$ and $1 - F = 3.145 \times 10^{-3}$ against -5.618758 and 3.143×10^{-3} for the full pool, at 11 408 against 23 312 CNOTs. At sixteen sites the single-layer ansatz has 1 022 parameters ($E = -7.197325$ against $E_0 = -7.209813$, $1 - F = 6.98 \times 10^{-3}$, 235 904 CNOTs) and its pruned version 255 parameters (7.00×10^{-3}, 61 584 CNOTs); all counts use the parity-support weight of the double excitations (Sec. 3.5).

ADAPT-VQE (Sec. 3.5) is run with gradient tolerance 10^{-6} and at most 200 operators, every amplitude re-optimized by L-BFGS warm-started from the previous iteration, with the exact ground state used only to record the infidelity. At eight sites ($U = 4t$, pool of 198) the infidelity falls below 10^{-2}, 3×10^{-3}, 10^{-3} and 3×10^{-4} after 66, 101, 149 and 194 operators (8 832, 11 672, 17 168 and 22 608 CNOTs including the Givens network; 2.77×10^{-4} after 200); at sixteen sites (pool of 1 022) it falls below 10^{-2} after 187 operators (48 008 CNOTs) and reaches 9.45×10^{-3} after 200. The CNOT count of a prefix is that of its operators as compiled in Sec. 5. The eight-site state at 149 operators, fed to the Trotterized filter of Sec. 5.3 (exact quantized-time ensemble), gives $1 - \mathcal{F}_M = 9.8 \times 10^{-4}$, 5.2×10^{-4}, 2.9×10^{-4}, 1.7×10^{-4}, 1.2×10^{-4}, 8.7×10^{-5} and 7.2×10^{-5} for $M = 0$–6 at 1.7×10^4, 3.7×10^4, 5.6×10^4, 7.6×10^4, 9.5×10^4, 1.1×10^5 and 1.3×10^5 CNOTs per accepted preparation.

S5 Trotterized refinement at eight sites

Figure S3 replaces the exact slice propagation at $T = 16/t$, $U = 4t$ by first- and second-order product formulas over the groups $[P(H_{\mathrm{low}} - \mu)P^\dagger, K_{\mathrm{even}}, K_{\mathrm{odd}}, D - \mu]$ for Path A and $[(1 + \Lambda \cos^2)K_{\mathrm{even}}, K_{\mathrm{odd}}, D]$ for Path B. The second-order formula reproduces the exact-slice overlap to within 10^{-3} with $n_s = 128$ steps for Path A ($p = 0.99983$), 256 for Path B with $\Lambda = 8t$ (0.99213) and 512 for $\Lambda = 16t$ (0.99342); first-order formulas do not reach it within 10^{-3} even at 512 steps. The counted circuit of Path A does not exponentiate the embedded coarse term as one block: on a device it is the coarse Hubbard step, i.e. the three conditioned coarse groups $P(D_c - \mu)P^\dagger$, $PK^c_{\mathrm{even}}P^\dagger$ and $PK^c_{\mathrm{odd}}P^\dagger$, which do not commute. Repeating the second-order re-

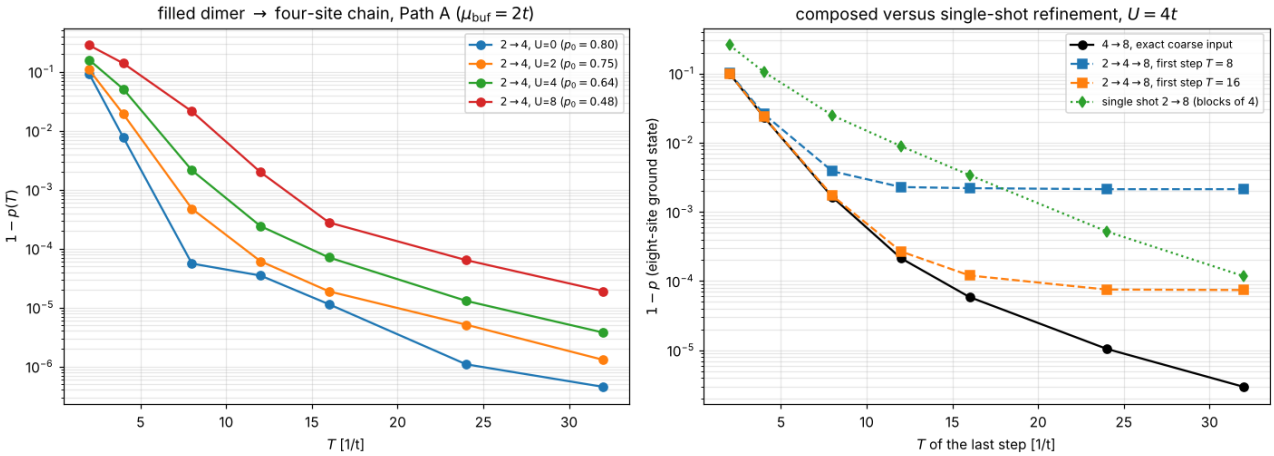


Figure S2: Left: $1-p(T)$ for the dimer step $2 \to 4$ at $U/t = 0, 2, 4, 8$. Right, $U = 4t$: the step $4 \to 8$ from the exact coarse ground state (circles), the composed chain $2 \to 4 \to 8$ with the refined four-site state as input after a first step of $T = 8/t$ or $16/t$ (squares), and the single-shot refinement $2 \to 8$ with blocks of four fine sites (diamonds), all versus the time of the last step.

Table S3: Plaquette refinement $2 \times 2 \to 4 \times 4$ on the open square lattice, Path A with $\mu_{\mathrm{buf}} = 2t$. At $(2, 2)$ the coarse ground state is a B_1 singlet with zero overlap on the A_2 triplet ground state of the fine lattice; the last row refines the lowest coarse $(A_2, S = 1)$ state, and Δ_{eff} is taken among the thirty lowest fine eigenstates.

$(N_\uparrow, N_\downarrow)$	U/t	dims	E_{low}	E_{high}	(Γ, S) coarse / fine	p_0	Δ	Δ_{eff}	$p(16)$
(1,1)	0	$16 \to 256$	-6.0000	-6.4721	A_1 / A_1	0.805	1.000	2.236	0.99999
(1,1)	4	$16 \to 256$	-5.8136	-6.2638	A_1 / A_1	0.840	0.792	2.096	0.99999
(1,1)	8	$16 \to 256$	-5.7093	-6.1833	A_1 / A_1	0.845	0.711	2.129	0.99998
(2,2)	4	$36 \to 14\,400$	-9.4142	-10.4197	$(B_1, 0)$ / $(A_2, 1)$	0	0.090	—	3×10^{-27}
(2,2)	4	$36 \to 14\,400$	-9.3429	-10.4197	$(A_2, 1)$ / $(A_2, 1)$	0.647	0.090	1.771	0.99997

finement with the coarse term split accordingly (six groups in all) gives $p = 0.997299$, 0.999818, 0.999941 and 0.999943 at $n_s = 64$, 128, 256 and 512, against 0.997456, 0.999831, 0.999941 and 0.999943 with the coarse term as one block; the step count $n_s = 128$ and the cost of Path A are unchanged, and the split-group state is the one used in Table 2. The ramp's product-formula version reproduces its exact-slice overlap with 64 second-order steps at $T = 8/t$ (0.996891 against 0.996861) and gives 0.999659 (exact 0.999885) with 128 steps at $T = 16/t$.

The embedded coarse term of Path A at eight sites is the coarse Hubbard step on the bonding register — 24 rotations, 32 CNOTs unconditioned — with every rotation conditioned on the eight antibonding qubits being empty, plus the conditioned scalar phase. With the Gray-code phase-polynomial convention of paper I (exact for few controls but growing as 2^{c+1}) this is 1.3×10^4 CNOTs per application; with a clean-ancilla ladder of relative-phase Toffolis per rotation 1 132; with the conjunction of the controls computed once into a shared flag, every rotation singly controlled by it and the flag uncomputed, 124, plus the frame change into and out of the bonding basis (32), i.e. 156 CNOTs per application, the figure used throughout (Sec. 5); the shared flag needs $c - 1 = 7$ clean work qubits. In the block Jordan–Wigner ordering the $Z_j Z_{j+L_f}$ terms are the only non-local ones; the site-major ordering, which makes them adjacent at the price of weight-three hopping strings, is the alternative to be compared on a linear-connectivity device.

S6 The exact-ensemble filter

Figure S4 shows the exact ensemble infidelity, Eq. (8), with $\sigma = 5/\Delta_{\mathrm{eff}}$ for the inputs of Sec. 4.5. Applied to the Path-B state at its ceiling ($\Lambda = 16t$, $T = 32/t$, $1 - p \simeq 2 \times 10^{-3}$) the filter reaches 2–4×10^{-5} after six cycles. The continuous-time values lie below the quantized-time, Trotterized ones of Table S5 once the latter approach the Trotter floor, which Eq. (8) does not contain (for the ramp state at $T = 16/t$, $n_s = 128$: 5.3×10^{-6}

Table S4: Sector-level UCCSD on the Galerkin coarse chain ($L = 4$, Eq. (4)) and on the fine chain ($L = 8$) at $(2, 2)$: reference, variational and exact energies, infidelity with the exact ground state, number of parameters and CNOT count of the preparation (Givens network plus excitation blocks). The last columns give the overlap of the prolonged UCCSD coarse state and its Path-A refinement at $T = 16/t$.

	coarse $L = 4$ (26 parameters, 1464 CNOTs)				fine $L = 8$ (198 parameters, 23312 CNOTs)				UCCSD coarse → refined	
U/t	E_{HF}	E_{UCCSD}	E_{FCI}	$1-F$	E_{HF}	E_{UCCSD}	E_{FCI}	$1-F$	p_0	$p(16)$
0	−6.23607	−6.23607	−6.23607	$< 10^{-15}$	−6.82295	−6.82295	−6.82295	$< 10^{-14}$	0.77733	0.99997
2	−5.23607	−5.43744	−5.43797	1.8×10^{-4}	−5.71184	−6.02962	−6.03209	4.7×10^{-4}	0.81405	0.99977
4	−4.23607	−4.96957	−4.97657	3.3×10^{-3}	−4.60073	−5.61876	−5.63443	3.1×10^{-3}	0.79991	0.99669
8	−2.23607	−4.51765	−4.55859	1.7×10^{-2}	−2.37850	−5.22167	−5.27797	1.2×10^{-2}	0.75667	0.98234

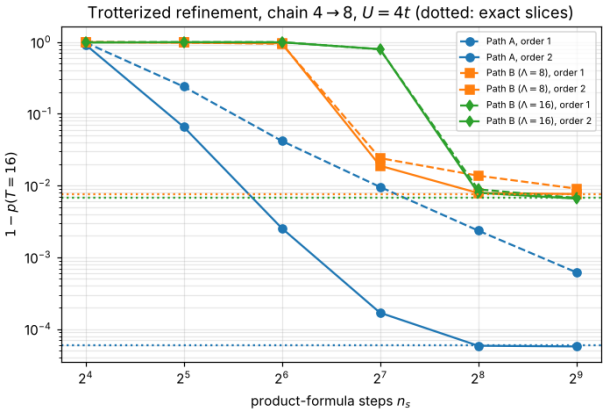


Figure S3: Trotterized refinement of the chain $4 \to 8$ at $U = 4t$, $T = 16/t$: $1 - p$ versus the number of product-formula steps for Path A and Path B ($\Lambda = 8t$, $16t$), first (dashed) and second order (solid); dotted lines are the exact-slice values.

against 6.2×10^{-5} at $M = 6$).

S7 Ingredients of the thirty-two-qubit study

Product formulas through the Kronecker structure. In the $(2, 2)$ sector of the sixteen-site chain (14 400 states) the dense machinery of the eight-site study is unavailable: a $14\,400^2$ complex step unitary cannot be stored, let alone diagonalized. Every term of the product formula is instead applied through Eq. (2). A hopping layer $K_{\mathrm{layer}} = K_\uparrow \otimes \mathbb{1} + \mathbb{1} \otimes K_\downarrow$ acts on ψ reshaped as a 120×120 matrix as $e^{-icK_\uparrow}\, \Psi\, e^{-icK_\downarrow^T}$, with the two 120×120 exponentials obtained from one diagonalization per species; the diagonal interaction is an elementwise phase; and the embedded coarse term of Path A, $P\,X\,P^\dagger$ with X one of the three 784×784 conditioned coarse groups, is applied exactly as $\psi \mapsto \psi + P[V(e^{-ic\lambda} - 1)V^\dagger]P^\dagger\psi$, with $X = V\lambda V^\dagger$ diagonalized once. The second-order step is the symmetric product of these factors. The ramp uses the groups $[K_{\mathrm{even}}, K_{\mathrm{odd}}, D]$ with the interaction coefficient $\sin^2(\pi s/2)$, Path B the same groups with the even (intra-block) layer scaled by $1 + \Lambda\cos^2(\pi s/2)$, and Path A the six groups $[P(D_c - \mu)P^\dagger, PK^c_{\mathrm{even}}P^\dagger, PK^c_{\mathrm{odd}}P^\dagger, K_{\mathrm{even}}, K_{\mathrm{odd}}, D - \mu]$ with the $\cos^2/\sin^2$ coefficients of Eq. (6). The exact-slice transports use 48 slices with a sparse Krylov exponential. Applied to the eight-site chain, where the dense machinery exists, the Kronecker implementation reproduces the ramp, the split-group Path-A refinement and the filter tracks of Sec. 5.3 to 10^{-13}.

The Trotter-effective ground state. The fixed-step filter projects, in the long run, on the eigenvector of the step unitary $U_{\delta t}$ closest to the ground state, and its eigenphase is the objective energy E_{obj}. At eight sites that eigenvector was read off a Schur decomposition of the step; at sixteen sites it is obtained by iterating the quantized-time filter itself: starting from the exact ground state, $\psi \mapsto \frac{1}{2}(\psi + e^{iE_{\mathrm{obj}}n\delta t}U^n_{\delta t}\psi)$ with n drawn from the Gaussian ensemble of cycle times and E_{obj} updated at every iteration to the eigenphase estimate $-\arg\langle\psi|U_{\delta t}|\psi\rangle/\delta t$, converges in a few tens of iterations to the fixed point of the filter, which is that eigenvector. At eight sites the procedure reproduces the Schur result ($E_{\mathrm{obj}} = -5.620907$, floor 5.8×10^{-5}) to the quoted digits; at sixteen sites, with $\sigma = 36.2/t$ and $\delta t = 0.1/t$, it gives $E_{\mathrm{obj}} - E_0 = +4.92 \times 10^{-3}t$ and a floor of 3.7×10^{-5}, close to the eight-site floor at the same step, as expected for a local error per step that is the same at both sizes.

Gate counts at sixteen sites. The second-order step of the sixteen-site chain costs 304 CNOTs (608 controlled, the scalar single-Z sum

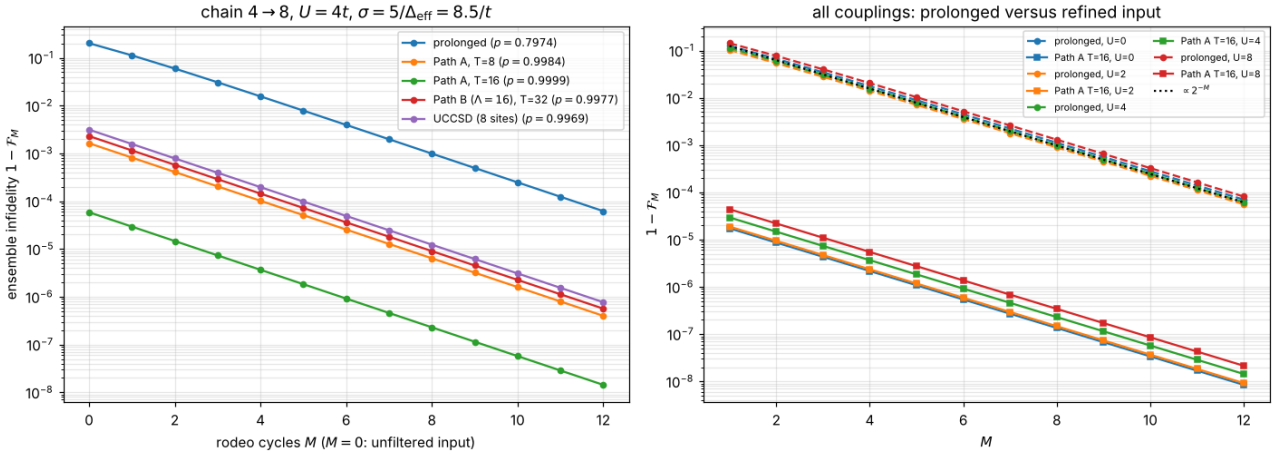


Figure S4: Exact ensemble infidelity of the rodeo filter, Eq. (8), with $\sigma = 5/\Delta_{\text{eff}}$ versus the number of cycles. Left, $U = 4t$: prolonged input, Path-A inputs refined for $T = 8/t$ and $16/t$, the Path-B input at its ceiling and the eight-site UCCSD state. Right: prolonged (dashed) and Path-A $T = 16/t$ (solid) inputs for all four couplings against the 2^{-M} law.

absorbed into the ancilla phase), the Givens network of the free-fermion determinant 2×120 rotations, 480 CNOTs, the prolongation $8 \to 16$ 32 CNOTs, and the coarse term of Path A — the eight-site Hubbard step, 52 rotations, conditioned on the sixteen antibonding qubits through one shared flag, plus the conditioned scalar phase and the frame change — $268 + 64 = 332$ CNOTs per application (4 948 with a relative-phase Toffoli ladder per rotation, the convention of an earlier version), applied twice per second-order step. The coarse state of the hierarchy is not an oracle: it is the state actually prepared by the cheapest eight-site unfiltered route of Sec. 5.3 below 3×10^{-4} (the ramp $T = 16/t$, $n_s = 192$: 1.8×10^{-4}, 27 760 CNOTs) or below 10^{-4} ($T = 24/t$, $n_s = 256$: 5.6×10^{-5}, 36 976 CNOTs), prolonged ($p_0 = 0.8689$ and 0.8688 against 0.8692) and transported, and its cost is charged to every input that starts from it; the search of Table 3 picks between the two, and the cheaper one wins at every target.

Step scans. With the step scans of the notebook (second order, $n_s = 32$ to 512, accepted when the overlap agrees with the exact-slice value to 10^{-3}): the ramp needs $n_s = 128$ at $T = 16/t$ (0.96927 against 0.96970; 0.91802 at $n_s = 64$) and 256 at $T = 32/t$ (0.99556 against 0.99591; 0.00545, 0.94463 and 0.99304 at $n_s = 32$, 64, 128 — a step of $\delta t = 1/t$ destroys the transport); Path B with $\Lambda = 8t$ needs 256 at $T = 16/t$ (0.99476 against 0.99492; 4×10^{-5}, 0.266 and 0.98688 at $n_s = 32$, 64, 128, the intra-block generator of norm $1 + \Lambda$ requiring $\delta t \lesssim 0.06/t$); Path A with $\mu_{\text{buf}} = 2t$ and the coarse groups split needs 64 steps at $T = 8/t$ (0.99228 against 0.99237; 0.99047 at $n_s = 32$) and 128 at $T = 16/t$ (0.99930 against 0.99931; 0.94602 and 0.99744 at $n_s = 32$, 64). An earlier version of this study reported a plateau of Path A at $T = 8/t$, 1.6×10^{-3} below the exact value at every step count, and explained it by the splitting of non-commuting coarse groups; the explanation was wrong. The embedded path had been built with the Galerkin constant $-tN_c$ instead of $-tN$ (Eq. (4)), i.e. with an extra term $-4t \cos^2(\pi s/2) P P^\dagger$ that is not a global phase, and with the correct constant the product formula converges to the exact-slice result at every T, as it must for a consistent decomposition.

Sector gap along the ramp. The sector gap of $H_R(s)$ falls monotonically from $0.165t$ at $s = 0$ to $0.051t$ at $s = 1$ ($0.125t$, $0.078t$, $0.057t$ at $s = 0.25, 0.5, 0.75$); the ramp from the determinant is governed, like Path A, by the gap in the singlet, reflection-even sector, which is why both transports converge at all with a sector gap of $0.05t$.

S8 Operating points

Table S5 lists, for a selection of the inputs that enter the first eight-site search behind Table 2, the exact quantized-time ensemble infidelity and the

CNOTs per accepted preparation after $M = 0$, 2, 4 and 6 cycles (that search ranged over 281 inputs, among them 42 ramps, 72 Path-B and 24 Path-A states from both charged coarse states and every ADAPT prefix below 2×10^{-2}; the refined search that gives the table's minima is described in Sec. S9); the sixteen-site data behind Table 3, with the bootstrap ranges of the minima, are given in Table S6.

S9 Convergence of the operating-point search

The minimum costs of Tables 2 and 3 of the main text are the best costs found under a stated finite search, and this section records how they move as the search is refined (notebook, Sec. 14). Three levels are compared. L0 is the grid of the first joint search (notebook, Sec. 13): at eight sites the ramp over $T \in \{8, 12, 16, 24, 32, 48\}/t$ and $n_s \in \{64, 96, \ldots, 512\}$, Path A over $T \in \{8, 12, 16\}/t$ and $n_s \in \{64, \ldots, 512\}$, Path B over $T \in \{16, 24, 32\}/t$ and $n_s \in \{128, \ldots, 1\,024\}$; at sixteen sites the ramp over $T \in \{32, 48, 64, 128\}/t$ and $n_s \in \{256, 512, 1\,024\}$ and Path A over six (T, n_s) pairs from $(8, 64)$ to $(32, 512)$. L1 is a fine grid of the evolution time — at eight sites $T = 6$–$48/t$ (ramp) and 6–$24/t$ (Path A) in steps of $1/t$ and 12–$32/t$ (Path B) in steps of $2/t$; at sixteen sites 16–$80/t$ in steps of $4/t$ (ramp) and 4–$40/t$ in steps of $2/t$ (Path A) — with the second-order step count on a geometric list of ratio 1.15 (32–768, 64–1 024 for Path B, 64–768 and 24–512 at sixteen sites) and, for Path A and B, both charged coarse states and $\Lambda \in \{8, 16, 32\}t$; at eight sites every point is evaluated with the exact quantized-time ensemble for $M = 0, \ldots, 6$, at sixteen sites unfiltered ($M = 0$), which is complete for the ramp and Path A because the expected cost of any filtered point is at least its input plus one average cycle, 1.76×10^5 CNOTs, above their unfiltered minima at every target of Table 3. L2 refines, for every control whose L1 cost lies within 40% of the L1 minimum, the integer step count by bracketing and bisection at fixed (T, M) and control. The coarse state of the $8 \to 16$ hierarchy is re-selected at L1 as the cheapest refined eight-site unfiltered ramp below 3×10^{-4} (c: $T = 13/t$, $n_s = 113$, $1 - F = 3.0 \times 10^{-4}$, 16 416 CNOTs with the prolongation, against 27 792 in the first search) and below 10^{-4} (a: $T = 19/t$, $n_s = 167$, 9.9×10^{-5}, 24 192 against 37 008). Table S7 lists the three levels, the L2 operating point, the number of distinct controls (T, or (Λ, T) with the coarse state) whose refined cost lies within 10% of the L2 minimum, and the ratio Path A/ramp. Figures S5 and S6 show the L0 grids themselves with the exact-ensemble or sampled tracks of the fixed inputs; the frontier curves of Fig. 6 of the main text are read from the L1 clouds.

Three features of Table S7 carry the conclusions drawn in the main text. First, the first grids were not converged: at eight sites every adiabatic minimum falls on refinement, by 28–44% for the ramp, 20–22% for Path A and 15–23% for Path B, and at sixteen sites by 25–63% for the ramp and 18–52% for Path A, the largest changes at the loose targets, where the first grid's shortest time and smallest step count were already too long and too many. Second, the fine grids are close to converged in the sense available to a finite search: the integer refinement L1→L2 moves the minima by 0–11%, and the unfiltered frontiers are sharp in T (one to six controls within 10% of the minimum) because the cheapest operating point sits at the shortest time at which the adiabatic error alone meets the target, after which the step count is fixed by the digitization error; the filtered Path-B frontier at eight sites is flatter (5–14 controls within 10%), as cycles and steps trade against each other there. We do not interpret differences between routes below the 10% level. Third, the ratio of the projector path to the ramp is 2.95–3.53 at eight sites and 1.60–1.86 at sixteen, uniformly in the ramp's favor: the first grid's near-tie at thirty-two qubits (1.5 against 1.6×10^5 CNOTs at 10^{-3}) was an artifact of its T spacing — the ramp's minimum moved from $(48/t, 512)$ to $(44/t, 244)$ and halved — and the fifteen-percent margin at 10^{-2} likewise ($(32/t, 256)$ to $(28/t, 94)$). The cheaper coarse states selected by the refined eight-site search (16 416 and 24 192 CNOTs instead of 27 792 and 37 008) lower the Path-A costs by 1–2×10^4 CNOTs and are included in the L1 and L2 values; the Path-B and prolonged-state rows of Table 3, which keep the first search's coarse states, would gain the same amount and remain far from the frontier.

Table S5: Eight-site inputs with the exact quantized-time ensemble ($\delta t = 0.1/t$, $\sigma = 8.5/t$, $E_{\mathrm{obj}} = E_{\mathrm{eff}}$): overlap p, preparation cost (the four-site coarse preparation of 8 216 CNOTs included for the inputs that start from the coarse state, marked c), ensemble infidelity and CNOTs per accepted preparation after $M = 0$, 2, 4 and 6 cycles.

input	p	prep	$1 - \mathcal{F}_M$, $M = 0, 2, 4, 6$				CNOTs per accepted preparation, $M = 0, 2, 4, 6$			
determinant	0.7804	112	2.2×10^{-1}	6.6×10^{-2}	1.7×10^{-2}	4.4×10^{-3}	1.1×10^{2}	4.5×10^{4}	8.7×10^{4}	1.3×10^{5}
prolonged (c)	0.7954	8 232	2.0×10^{-1}	6.0×10^{-2}	1.6×10^{-2}	4.0×10^{-3}	8.2×10^{3}	5.4×10^{4}	9.6×10^{4}	1.4×10^{5}
UCCSD pruned (98)	0.9969	11 408	3.1×10^{-3}	8.3×10^{-4}	2.5×10^{-4}	1.1×10^{-4}	1.1×10^{4}	5.1×10^{4}	9.0×10^{4}	1.3×10^{5}
UCCSD full (198)	0.9969	23 312	3.1×10^{-3}	8.3×10^{-4}	2.5×10^{-4}	1.1×10^{-4}	2.3×10^{4}	6.3×10^{4}	1.0×10^{5}	1.4×10^{5}
ADAPT (149 op.)	0.9990	17 168	9.8×10^{-4}	2.9×10^{-4}	1.2×10^{-4}	7.2×10^{-5}	1.7×10^{4}	5.6×10^{4}	9.5×10^{4}	1.3×10^{5}
ADAPT (194 op.)	0.9997	22 608	3.0×10^{-4}	1.2×10^{-4}	7.3×10^{-5}	6.1×10^{-5}	2.3×10^{4}	6.2×10^{4}	1.0×10^{5}	1.4×10^{5}
ramp, $T = 8$, $n_s = 64$	0.9969	9 328	3.1×10^{-3}	8.2×10^{-4}	2.5×10^{-4}	1.1×10^{-4}	9.3×10^{3}	4.9×10^{4}	8.8×10^{4}	1.3×10^{5}
ramp, $T = 12$, $n_s = 96$	0.9994	13 936	5.9×10^{-4}	1.9×10^{-4}	9.0×10^{-5}	6.6×10^{-5}	1.4×10^{4}	5.3×10^{4}	9.2×10^{4}	1.3×10^{5}
ramp, $T = 16$, $n_s = 128$	0.9997	18 544	3.4×10^{-4}	1.3×10^{-4}	7.5×10^{-5}	6.2×10^{-5}	1.9×10^{4}	5.8×10^{4}	9.7×10^{4}	1.4×10^{5}
ramp, $T = 16$, $n_s = 192$	0.9998	27 760	1.8×10^{-4}	8.8×10^{-5}	6.5×10^{-5}	6.0×10^{-5}	2.8×10^{4}	6.7×10^{4}	1.1×10^{5}	1.5×10^{5}
ramp, $T = 24$, $n_s = 256$	0.99994	36 976	5.6×10^{-5}	5.7×10^{-5}	5.7×10^{-5}	5.8×10^{-5}	3.7×10^{4}	7.6×10^{4}	1.2×10^{5}	1.5×10^{5}
Path B ($\Lambda = 8t$), $T = 16$, $n_s = 256$ (c)	0.9917	45 096	8.3×10^{-3}	2.1×10^{-3}	5.7×10^{-4}	1.9×10^{-4}	4.5×10^{4}	8.5×10^{4}	1.2×10^{5}	1.6×10^{5}
Path B ($\Lambda = 16t$), $T = 24$, $n_s = 512$ (c)	0.9967	81 960	3.4×10^{-3}	8.8×10^{-4}	2.6×10^{-4}	1.1×10^{-4}	8.2×10^{4}	1.2×10^{5}	1.6×10^{5}	2.0×10^{5}
Path A, $T = 12$, $n_s = 64$ (c)	0.9991	37 416	8.9×10^{-4}	2.7×10^{-4}	1.1×10^{-4}	7.1×10^{-5}	3.7×10^{4}	7.7×10^{4}	1.2×10^{5}	1.5×10^{5}
Path A, $T = 12$, $n_s = 128$ (c)	0.9997	66 600	2.7×10^{-4}	1.1×10^{-4}	7.2×10^{-5}	6.1×10^{-5}	6.7×10^{4}	1.1×10^{5}	1.4×10^{5}	1.8×10^{5}
Path A, $T = 16$, $n_s = 128$ (c)	0.9997	66 600	2.9×10^{-4}	1.2×10^{-4}	7.3×10^{-5}	6.2×10^{-5}	6.7×10^{4}	1.1×10^{5}	1.4×10^{5}	1.8×10^{5}

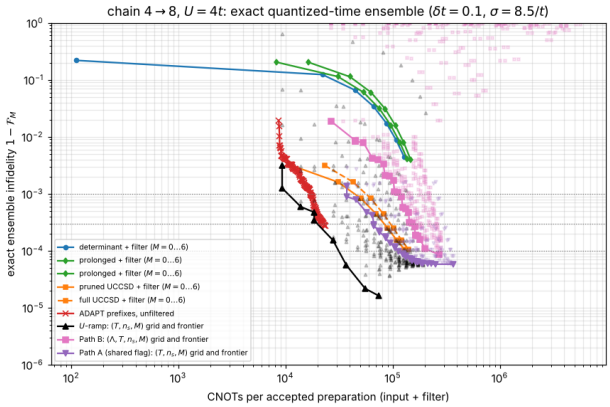


Figure S5: The L0 grid at eight sites with the exact quantized-time ensemble (chain $4 \to 8$, $U = 4t$): lines with markers are the fixed inputs as M grows from 0 to 6 and the unfiltered ADAPT prefixes; faint symbols are every point of the (T, n_s, M) grid of the ramp, the (Λ, T, n_s, M) grid of Path B and the (T, n_s, M) grid of Path A (both coarse states), with their lower envelopes drawn (the L0 minima of Table S7); dotted lines mark the targets of Table 2.

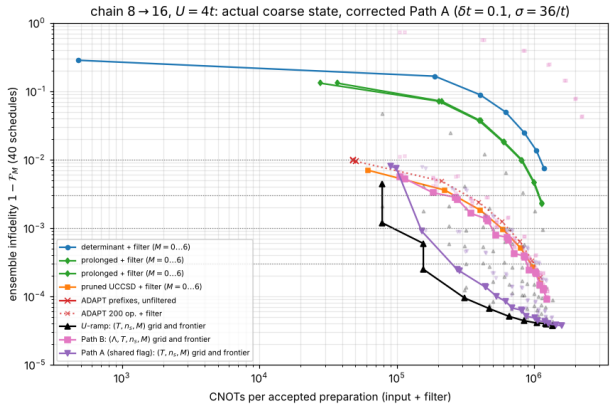


Figure S6: The L0 grid at sixteen sites (chain $8 \to 16$, $U = 4t$, forty shared time sets): the fixed inputs as M grows from 0 to 6, the unfiltered ADAPT prefixes and the 200-operator state through the filter, and every point of the ramp, Path-B and Path-A grids with their lower envelopes; dotted lines mark the targets of Table 3.

References

[1] S. Jansen, M.-B. Ruskai, and R. Seiler, J. Math. Phys. **48**, 102111 (2007).

Table S6: The sixteen-site search behind Table 3: the fixed inputs, the two charged coarse states (c: the eight-site ramp $T = 16/t$, $n_s = 192$, 27 760 CNOTs; a: $T = 24/t$, $n_s = 256$, 36 976 CNOTs) and the grid points that attain a minimum of their route, with overlap p, preparation cost (coarse preparation included), ensemble infidelity (bootstrap relative standard error over the forty shared time sets in parentheses) and CNOTs per accepted preparation after $M = 0$, 2, 4 and 6 cycles, and the cumulative acceptance after six. The search contains 47 inputs: the determinant, the two prolonged states, the pruned template, the ADAPT prefixes at 187, 190 and 200 operators, 12 ramps ($T = 32$–$128/t$, $n_s = 256$–$1\,024$), 16 Path-B states ($\Lambda = 8t$, $16t$; $T = 16$, $32/t$; $n_s = 256$, 512; both coarse states) and 12 Path-A states ($T = 8$–$32/t$, $n_s = 64$–512; both coarse states). In 400 bootstrap resamples of the schedules the 5–95% ranges of the minimum cost per route are $[7.8, 7.8]$, $[7.8, 7.8]$, $[16, 16]$, $[16, 16] \times 10^4$ for the ramp, $[9.0, 9.0]$, $[15, 15]$, $[15, 15]$, $[28, 28] \times 10^4$ for Path A, $[4.8, 4.8]$, $[35, 45]$, $[72, 87]$, $[95, 120] \times 10^4$ for ADAPT, $[6.2, 6.2]$, $[36, 46]$, $[54, 82]$, $[91, 110] \times 10^4$ for the pruned template and $[11, 11]$, $[23, 30]$, $[48, 58]$, $[84, 99] \times 10^4$ for Path B at the targets 10^{-2}, 3×10^{-3}, 10^{-3} and 3×10^{-4}.

input	p	prep	$M = 0$		$M = 2$		$M = 4$		$M = 6$		
			$1 - \mathcal{F}$	CNOTs	$1 - \mathcal{F}$	CNOTs	$1 - \mathcal{F}$	CNOTs	$1 - \mathcal{F}$	CNOTs	P_6
determinant	0.7162	480	2.8×10^{-1}	4.8×10^{2}	8.9×10^{-2} (4%)	4.1×10^{5}	2.5×10^{-2} (7%)	8.4×10^{5}	7.4×10^{-3} (14%)	1.2×10^{6}	0.717
prolonged (c)	0.8689	27 792	1.3×10^{-1}	2.8×10^{4}	3.7×10^{-2} (4%)	4.0×10^{5}	9.8×10^{-3} (11%)	8.1×10^{5}	2.3×10^{-3} (10%)	1.1×10^{6}	0.870
prolonged (a)	0.8688	37 008	1.3×10^{-1}	3.7×10^{4}	3.7×10^{-2} (4%)	4.1×10^{5}	9.8×10^{-3} (10%)	8.2×10^{5}	2.3×10^{-3} (10%)	1.1×10^{6}	0.870
UCCSD pruned (255)	0.9930	61 584	7.0×10^{-3}	6.2×10^{4}	1.8×10^{-3} (7%)	4.1×10^{5}	5.1×10^{-4} (11%)	8.0×10^{5}	1.6×10^{-4} (15%)	1.1×10^{6}	0.993
ADAPT (187 op.)	0.9901	48 008	9.9×10^{-3}	4.8×10^{4}	2.5×10^{-3} (6%)	3.9×10^{5}	6.6×10^{-4} (10%)	7.9×10^{5}	2.0×10^{-4} (13%)	1.1×10^{6}	0.990
ADAPT (200 op.)	0.9906	50 392	9.5×10^{-3}	5.0×10^{4}	2.4×10^{-3} (6%)	4.0×10^{5}	6.3×10^{-4} (9%)	7.9×10^{5}	1.9×10^{-4} (14%)	1.1×10^{6}	0.990
ramp, $T = 32$, $n_s = 256$	0.9956	78 304	4.4×10^{-3}	7.8×10^{4}	1.1×10^{-3} (12%)	4.2×10^{5}	2.7×10^{-4} (14%)	8.1×10^{5}	9.6×10^{-5} (22%)	1.1×10^{6}	0.996
ramp, $T = 48$, $n_s = 256$	0.9988	78 304	1.2×10^{-3}	7.8×10^{4}	3.1×10^{-4} (9%)	4.2×10^{5}	1.1×10^{-4} (10%)	8.1×10^{5}	5.6×10^{-5} (11%)	1.1×10^{6}	0.999
ramp, $T = 48$, $n_s = 512$	0.9994	156 128	5.9×10^{-4}	1.6×10^{5}	1.6×10^{-4} (12%)	5.0×10^{5}	6.7×10^{-5} (8%)	8.9×10^{5}	4.4×10^{-5} (7%)	1.2×10^{6}	0.999
ramp, $T = 64$, $n_s = 512$	0.99975	156 128	2.5×10^{-4}	1.6×10^{5}	8.5×10^{-5} (6%)	5.0×10^{5}	5.0×10^{-5} (4%)	8.9×10^{5}	4.0×10^{-5} (3%)	1.2×10^{6}	1.000
ramp, $T = 128$, $n_s = 1024$	0.99991	311 776	9.5×10^{-5}	3.1×10^{5}	5.1×10^{-5} (1%)	6.5×10^{5}	4.1×10^{-5} (1%)	1.0×10^{6}	3.8×10^{-5} (0%)	1.4×10^{6}	1.000
Path B ($\Lambda = 8t$), $T = 16$, $n_s = 256$ (c)	0.9943	105 616	5.7×10^{-3}	1.1×10^{5}	1.4×10^{-3} (4%)	4.5×10^{5}	3.9×10^{-4} (9%)	8.4×10^{5}	1.3×10^{-4} (10%)	1.2×10^{6}	0.994
Path B ($\Lambda = 16t$), $T = 32$, $n_s = 512$ (c)	0.9967	183 440	3.3×10^{-3}	1.8×10^{5}	7.9×10^{-4} (4%)	5.3×10^{5}	2.4×10^{-4} (8%)	9.2×10^{5}	9.1×10^{-5} (7%)	1.2×10^{6}	0.997
Path A, $T = 8$, $n_s = 64$ (c)	0.9921	89 744	7.9×10^{-3}	9.0×10^{4}	1.6×10^{-3} (14%)	4.3×10^{5}	5.3×10^{-4} (23%)	8.3×10^{5}	1.8×10^{-4} (23%)	1.1×10^{6}	0.992
Path A, $T = 16$, $n_s = 128$ (c)	0.9991	151 696	9.0×10^{-4}	1.5×10^{5}	2.2×10^{-4} (7%)	4.9×10^{5}	8.6×10^{-5} (9%)	8.9×10^{5}	5.0×10^{-5} (6%)	1.2×10^{6}	0.999
Path A, $T = 16$, $n_s = 128$ (a)	0.9991	160 912	9.2×10^{-4}	1.6×10^{5}	2.2×10^{-4} (7%)	5.0×10^{5}	8.9×10^{-5} (10%)	9.0×10^{5}	5.1×10^{-5} (7%)	1.2×10^{6}	0.999
Path A, $T = 32$, $n_s = 256$ (c)	0.99975	275 600	2.5×10^{-4}	2.8×10^{5}	8.4×10^{-5} (3%)	6.2×10^{5}	4.9×10^{-5} (2%)	1.0×10^{6}	4.0×10^{-5} (1%)	1.3×10^{6}	1.000
Path A, $T = 32$, $n_s = 512$ (a)	0.99990	532 624	1.0×10^{-4}	5.3×10^{5}	5.1×10^{-5} (2%)	8.8×10^{5}	4.0×10^{-5} (1%)	1.3×10^{6}	3.7×10^{-5} (1%)	1.6×10^{6}	1.000

Table S7: Convergence of the operating-point search: best CNOTs per accepted preparation found at the three search levels described in the text (L0: grid of the first joint search; L1: fine T grid, geometric n_s list; L2: L1 plus the integer refinement of n_s), the L2 operating point as (T, n_s, M) or (Λ, T, n_s, M) with the charged coarse state, the infidelity reached there, the number of distinct controls within 10% of the L2 minimum, and the ratio Path A/ramp at L2. Filter settings and costs as in Tables 2 and 3.

route	target	L0	L1	L2	L2 point	$1 - \mathcal{F}$	within 10%	A/ramp
chain $4 \to 8$, sixteen qubits (exact quantized-time ensemble)								
U-ramp	10^{-3}	1.40×10^4	1.13×10^4	1.00×10^4	(13, 69, 0)	9.6×10^{-4}	3	
	3×10^{-4}	2.80×10^4	1.74×10^4	1.57×10^4	(14, 108, 0)	3.0×10^{-4}	2	
	10^{-4}	3.70×10^4	2.65×10^4	2.42×10^4	(19, 167, 0)	9.9×10^{-5}	1	
Path A	10^{-3}	3.70×10^4	3.10×10^4	2.97×10^4	(9, 47, 0; c)	9.8×10^{-4}	1	2.95
	3×10^{-4}	6.70×10^4	5.57×10^4	5.34×10^4	(13, 99, 0; c)	3.0×10^{-4}	4	3.41
	10^{-4}	1.10×10^5	8.89×10^4	8.53×10^4	(18, 151, 0; a)	9.9×10^{-5}	2	3.53
Path B	10^{-3}	1.20×10^5	1.01×10^5	9.85×10^4	(8, 18, 216, 3; c)	1.0×10^{-3}	10	
	3×10^{-4}	1.60×10^5	1.36×10^5	1.36×10^5	(8, 14, 202, 5; c)	3.0×10^{-4}	14	
	10^{-4}	2.70×10^5	2.16×10^5	2.09×10^5	(16, 28, 577, 6; c)	1.0×10^{-4}	5	
chain $8 \to 16$, thirty-two qubits (unfiltered, $M = 0$)								
U-ramp	10^{-2}	7.8×10^4	3.06×10^4	2.91×10^4	(28, 94, 0)	9.9×10^{-3}	2	
	3×10^{-3}	7.8×10^4	4.67×10^4	4.43×10^4	(40, 144, 0)	3.0×10^{-3}	1	
	10^{-3}	1.6×10^5	8.20×10^4	7.47×10^4	(44, 244, 0)	1.0×10^{-3}	2	
	3×10^{-4}	1.6×10^5	1.25×10^5	1.19×10^5	(60, 391, 0)	3.0×10^{-4}	2	
Path A	10^{-2}	9.0×10^4	4.84×10^4	4.64×10^4	(8, 31, 0; c)	9.9×10^{-3}	3	1.60
	3×10^{-3}	1.5×10^5	7.35×10^4	7.16×10^4	(12, 57, 0; c)	3.0×10^{-3}	2	1.62
	10^{-3}	1.5×10^5	1.26×10^5	1.23×10^5	(18, 102, 0; a)	9.8×10^{-4}	6	1.65
	3×10^{-4}	2.8×10^5	2.31×10^5	2.22×10^5	(26, 204, 0; a)	3.0×10^{-4}	5	1.86